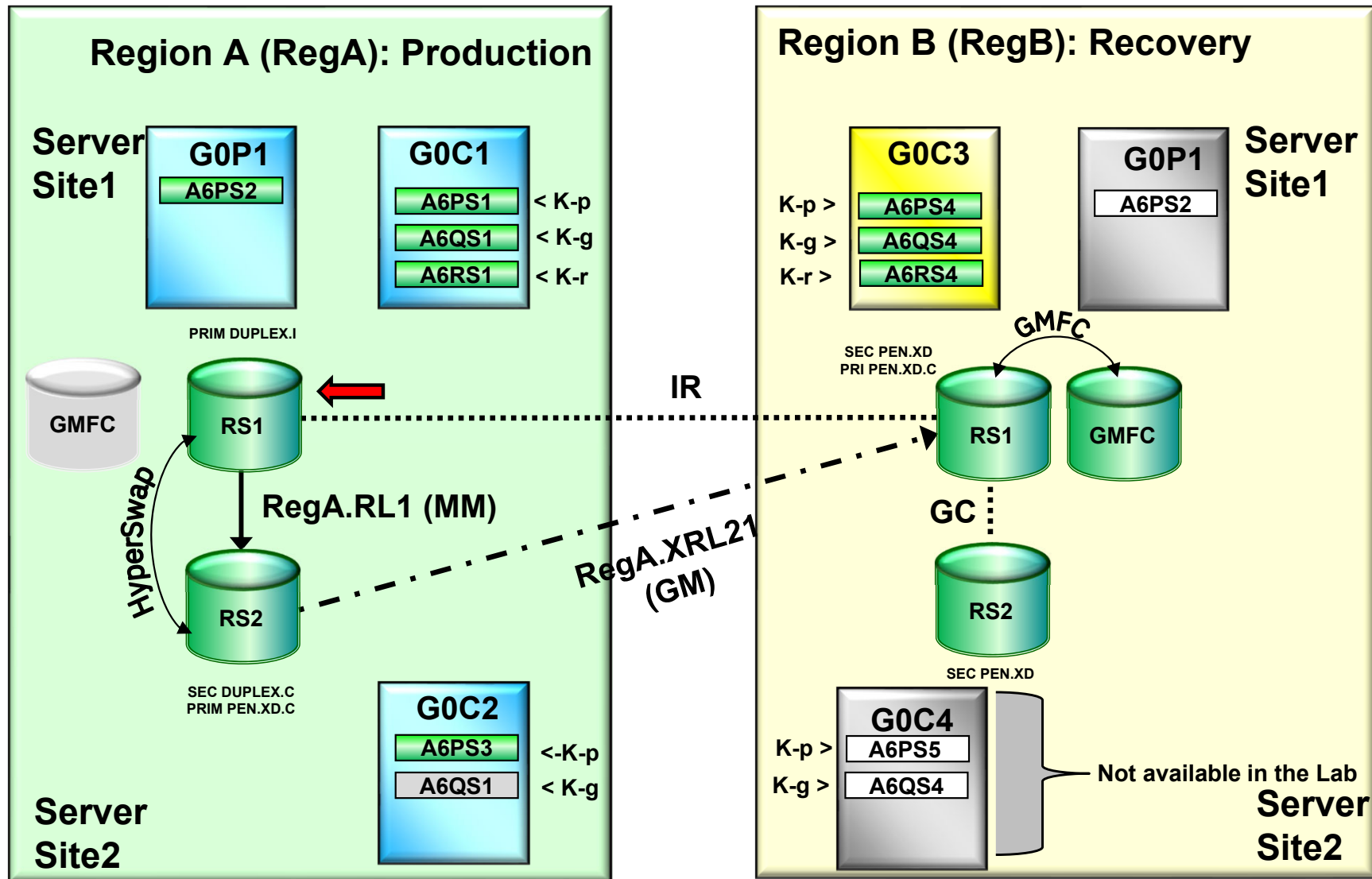


GDPS METRO GLOBAL – LAB: REGION SWITCH



Planned Region Switch

- with initial configuration MGM cascaded
 - Starting configuration



Planned region switch

- with initial configuration MGM cascaded and primary are RegA.RS1

■ High level steps

- Shutdown all production systems and the Master Kp-Sys
- *Shutdown remaining RegA Kp-Sys and reset LPAR
- *Deactivate/Activate RegA CFs
- *Reload Kp-Sys in RegA.SS1 in 'recovery' mode and start GDPS Netviews (Kp, Kg and Kr) and validate GDPS Site-Table and CDSs
- Run MGM4SITE_SWITCH_CAS_RS1 procedure on Kg-Sys, it will
 - Region switch RegA > RegB
- Initialize GM Session RegB.RS2 > RegA.RS1
- Optionally Stop Mirroring RegB.RS1 > RegB.RS2
- *Shutdown Kp-Sys in RegB.SS1 and LPAR reset
- *Deactivate/Activate RegB CFs
- *Reload RegB.SS1 Kp-Sys in 'production' mode and start GDPS Netviews (Kp, Kg and Kr) and validate GDPS Site-Table and CDSs
- Activate LPARs and load production systems from Kp-Sys in RegB
- Validate production systems in RegB
- After decision to stay in RegB restart Mirroring RegB.RS1 > RegB.RS2
- Execute MGM4SITE_CHECK_TOPOLOGY from RegB Kg-sys

*see additional considerations

Planned region switch

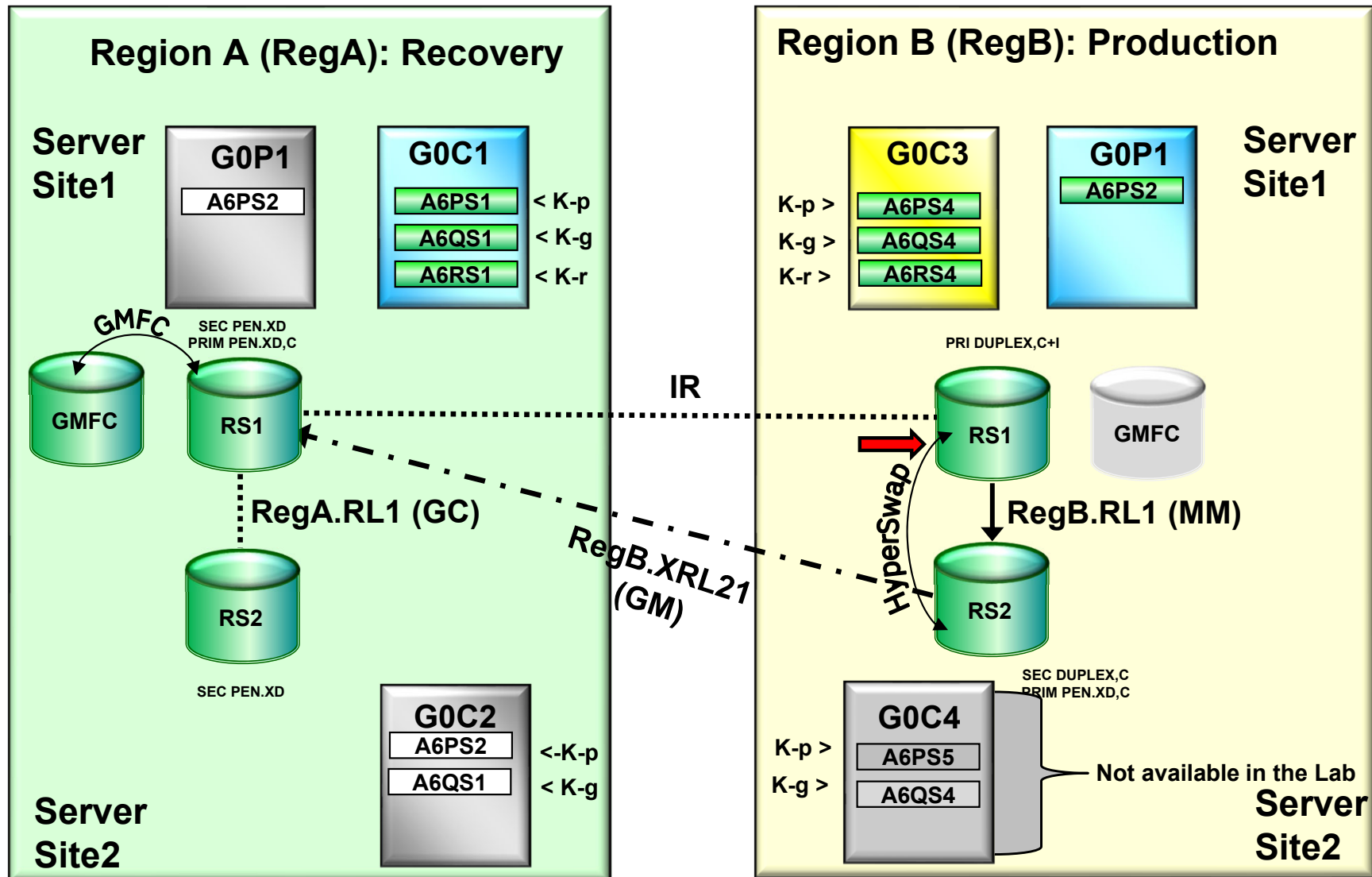
- with initial configuration MGM cascaded and primary are RegA.RS1

- Additional considerations

- *If running zOS 2.2 and ALLOWACCESS(NO) is specified to prevent Controlling System LOGR CDS allocation, there is no need to switch Controlling systems between 'production' and 'recovery' modes
- Cleanup of CF structures in the old application site may still be desired
- If a RegB.RS2 is being used for testing, it needs to be cleaned up by running a control script MGM4SITE_TESTCOPY_END prior to the SWITCH procedure
- Region Switch can only be performed in a cascaded configuration
 - Procedure MGM4SITE_SWITCH_CAS_RS2 is also provided

Planned Region Switch

- with initial configuration MGM cascaded
- Target configuration



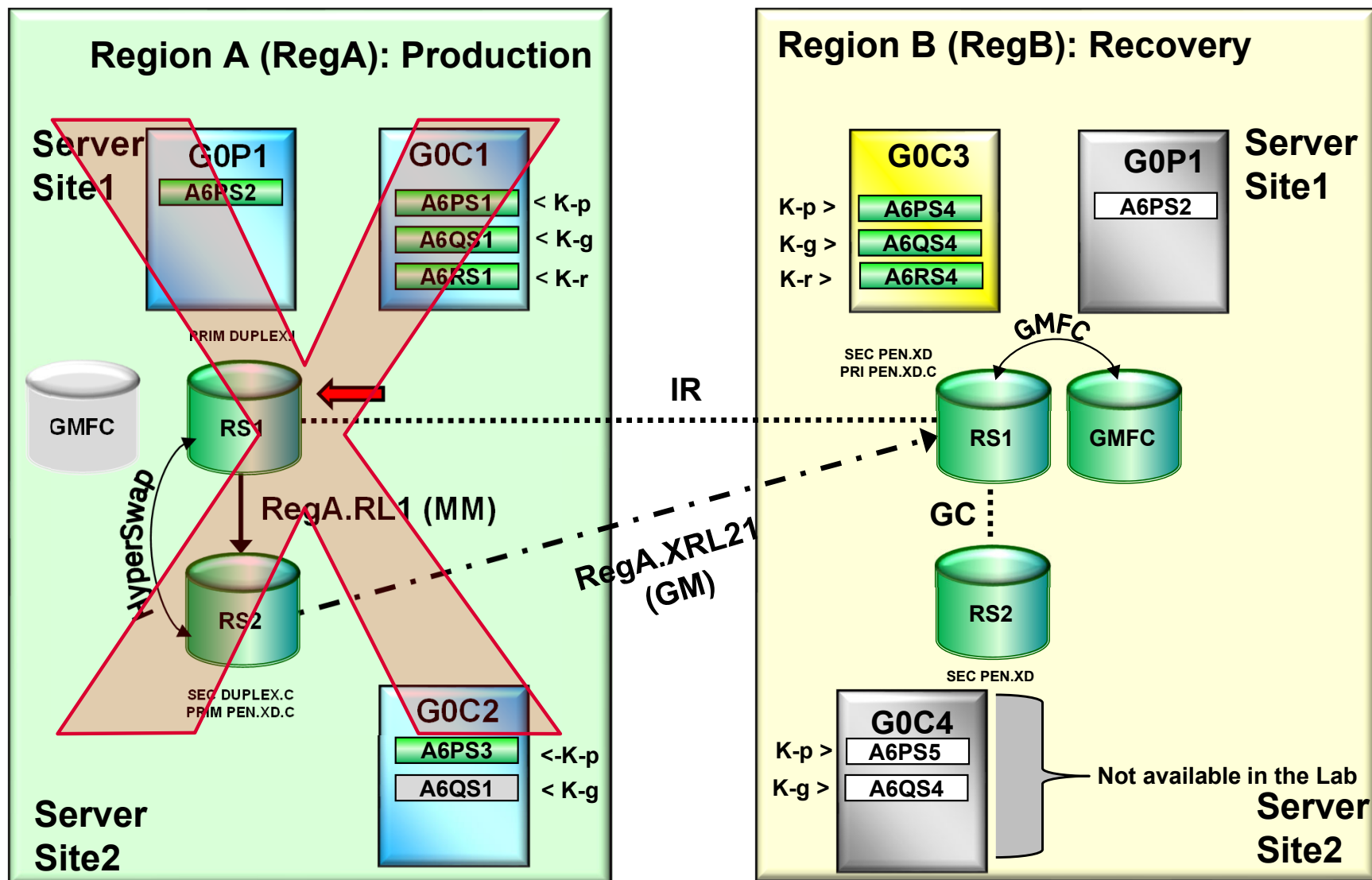
Planned region switch – return to normal

- to initial configuration MGM cascaded

- Current situation:
 - RegB.RS1 > RegB.RS2 (MM) > RegA.RS1 (GM) > RegA.RS2 (GC)
- High level procedure:
 - Simply a repeat of the previously discussed process

UnPlanned Region Switch

- with initial configuration MGM cascaded
 - Starting configuration



Unplanned region switch

- with initial configuration MGM cascaded

■ High level steps

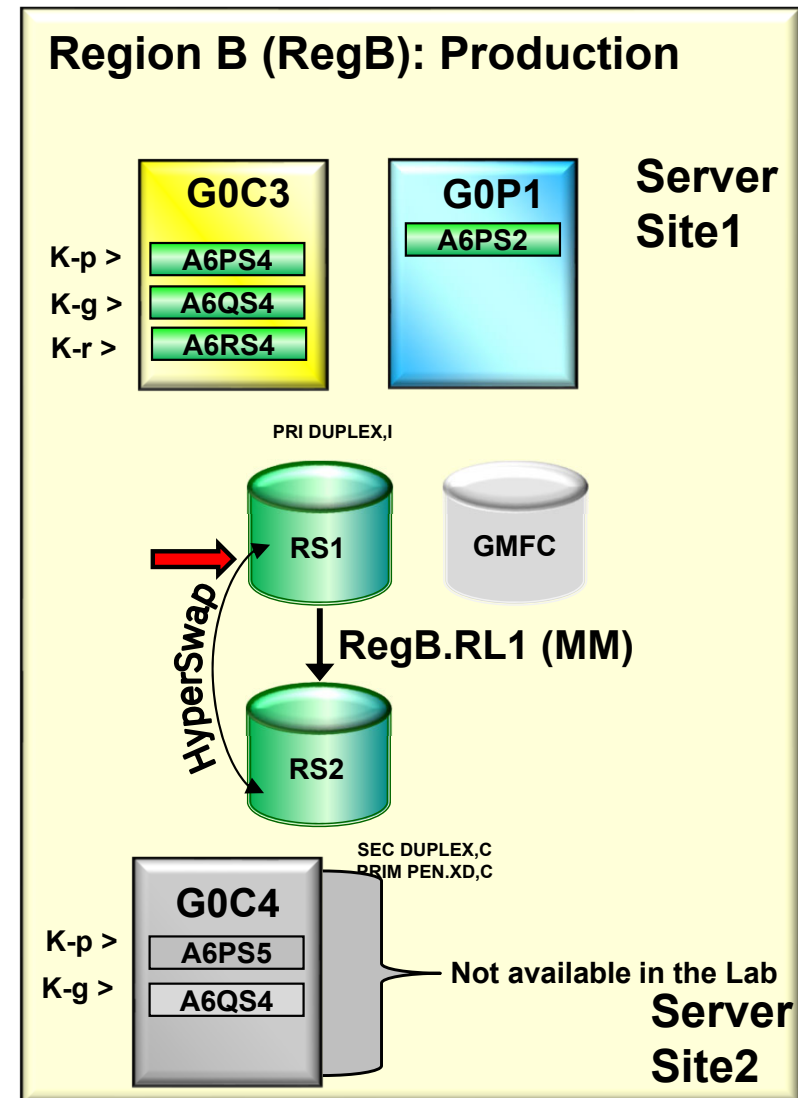
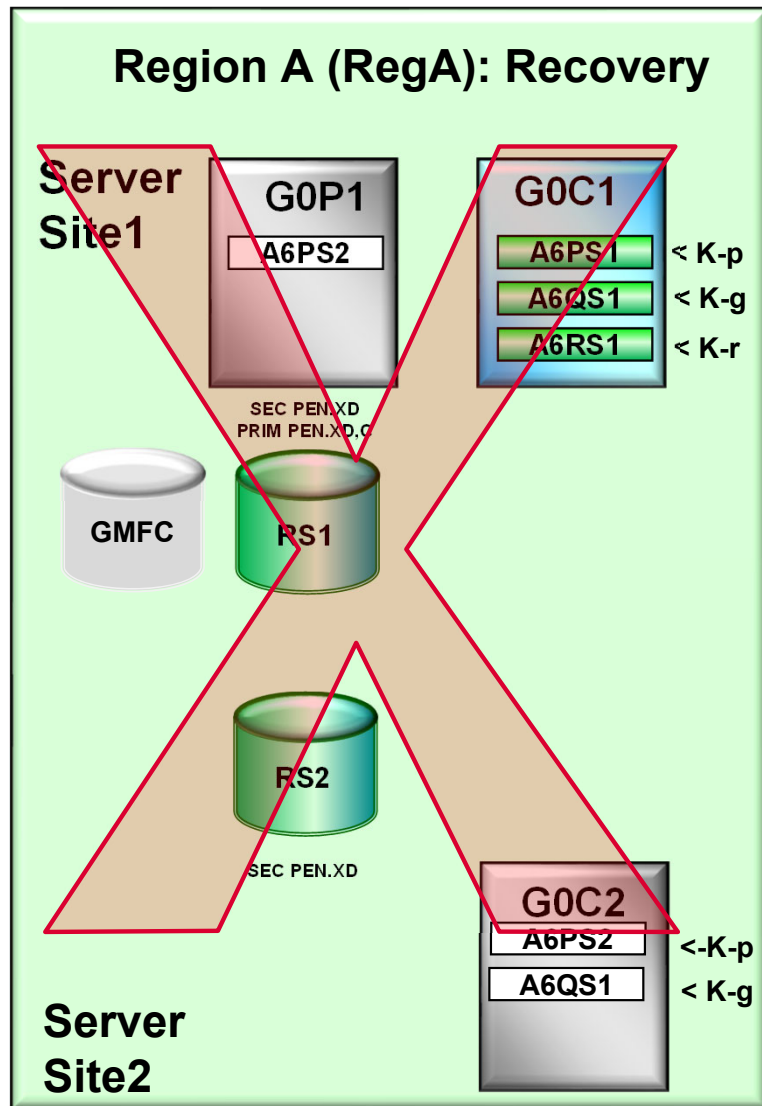
- Reset all production systems in RegA
 - Due to lost connection to Kp-Sys or HMC RegA it might not be possible
- Run control script with GDASD RECOVER from Kr-Sys in RegB
- Activate LPARs and load production systems from Kr-Sys in RegB
- Transition RegB.RS1 > RegB.RS2 to synchronous mode by executing DASD='START SECONDARY'

■ Additional consideration

- Since the FC relationship RegB.RS1 > RegB.GMFC still exists it could impact the host I/O of RegB.RS1 disks
- It is recommended to remove the FC relationship by executing a control script with GDASD CLEANUPR GMFC on Kr-Sys

Un Planned Region Switch

- with initial configuration MGM cascaded
- Target configuration



Unplanned region switch – preparing to return to normal

- to initial configuration MGM cascaded

- Current situation

- RegA.RS1 and RegA.RS2 are available again and still contain the data and metadata
- Systems in RegA are down
- No Copy Services relationship RegB.RS2 > RegA
- Production runs in RegB with no DR protection

- High level steps:

- From RegA HMC
 - reset all systems
 - deactivate and activate CFs (cleanup)
 - IPL G0C1 with Kp- and Kg-Instances

Unplanned region switch – preparing to return to normal

- to initial configuration MGM cascaded

▪ High level steps (continued):

- Run control scripts with DASD STOP SECONDARY and DASD START SECONDARY XD from RegA Kp-Sys
 - Will establish PPRC-XD in RegA from the site the Primary Disk site indicator is pointing to RS1
- Run procedure MGM4SITE_PREPARE_MIRROR from Kg-Sys, it will:
 - Evaluate the current state of cross region replication and if IR cannot be exploited will fail requiring manual clean up of residual PPRC-XD relationships
 - When executed successfully mirroring will be established RegB.RS1 > RegB.RS2 > either RegA.RS1 or RegA.RS2 with Global Copy between RegA sites.
- MGM4SITE_PREPARE_MIRROR_SWAP may need to be run prior to restarting the GM session

Unplanned region switch – preparing to return to normal

- to initial configuration MGM cascaded

▪ High level steps (continued):

- Run control scripts with DASD STOP SECONDARY and DASD START SECONDARY XD from RegA Kp-Sys
 - Will establish PPRC-XD in RegA from the site the Primary Disk site indicator is pointing to RS1
- Run procedure MGM4SITE_PREPARE_MIRROR from Kg-Sys, it will:
 - Evaluate the current state of cross region replication and if IR cannot be exploited will fail requiring manual clean up of residual PPRC-XD relationships
 - When executed successfully mirroring will be established RegB.RS1 > RegB.RS2 > either RegA.RS1 or RegA.RS2 with Global Copy between RegA sites.
- MGM4SITE_PREPARE_MIRROR_SWAP may need to be run prior to restarting the GM session

Unplanned region switch – preparing to return to normal

- to initial configuration MGM cascaded

- High level steps (continued):

- MGM4SITE_RESTORE may need to be run prior to starting region switch
- MGM4SITE_SWITCH

Unplanned region switch – preparing to return to normal

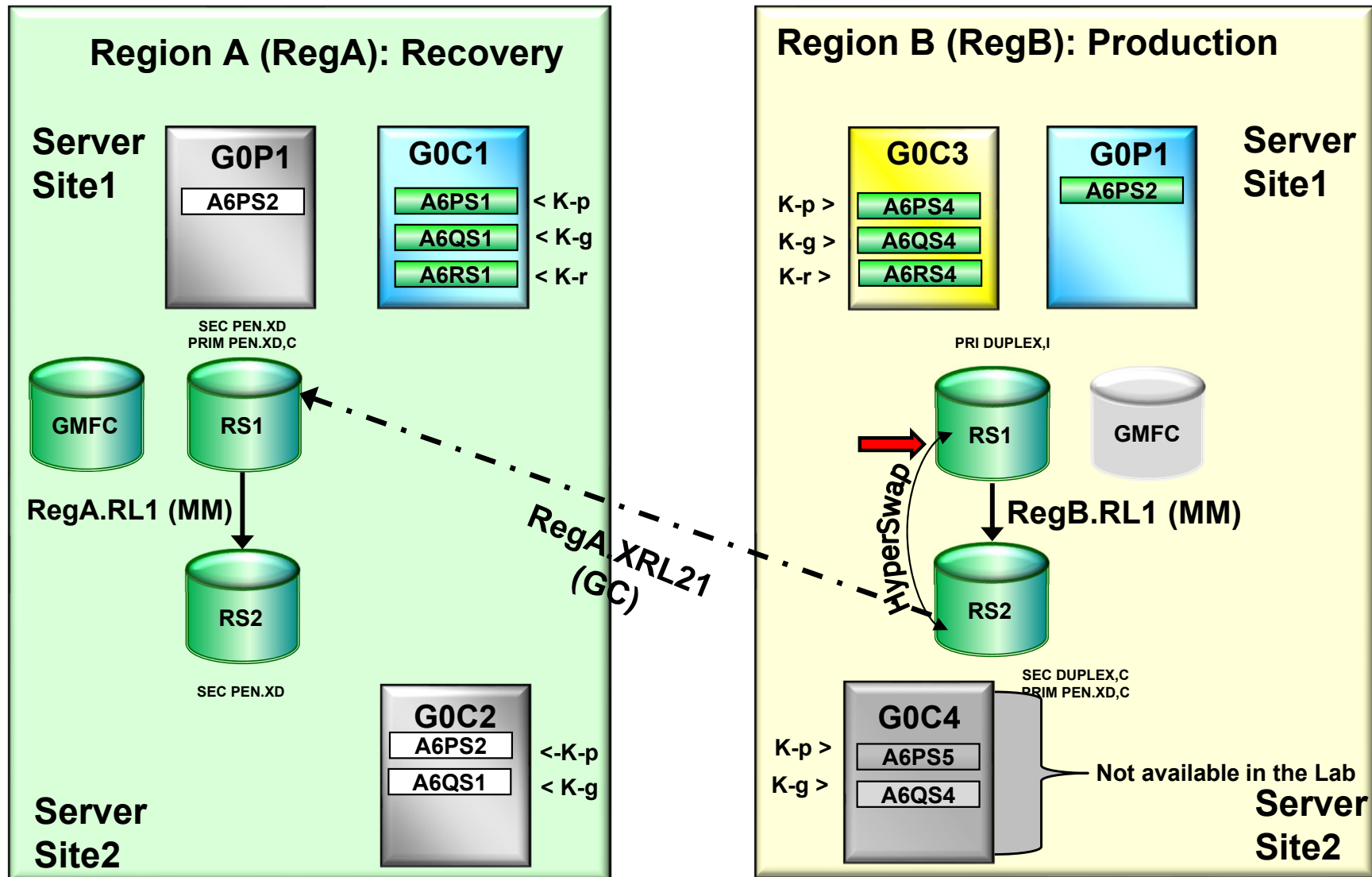
- to initial configuration MGM cascaded

- High level steps (continued):

- Restore recovery site readiness by re-initialising the GM session from RegB to RegA.
- Once all replication is in place use the MGM4SITE_SWITCH_CASx procedure to return to RegA

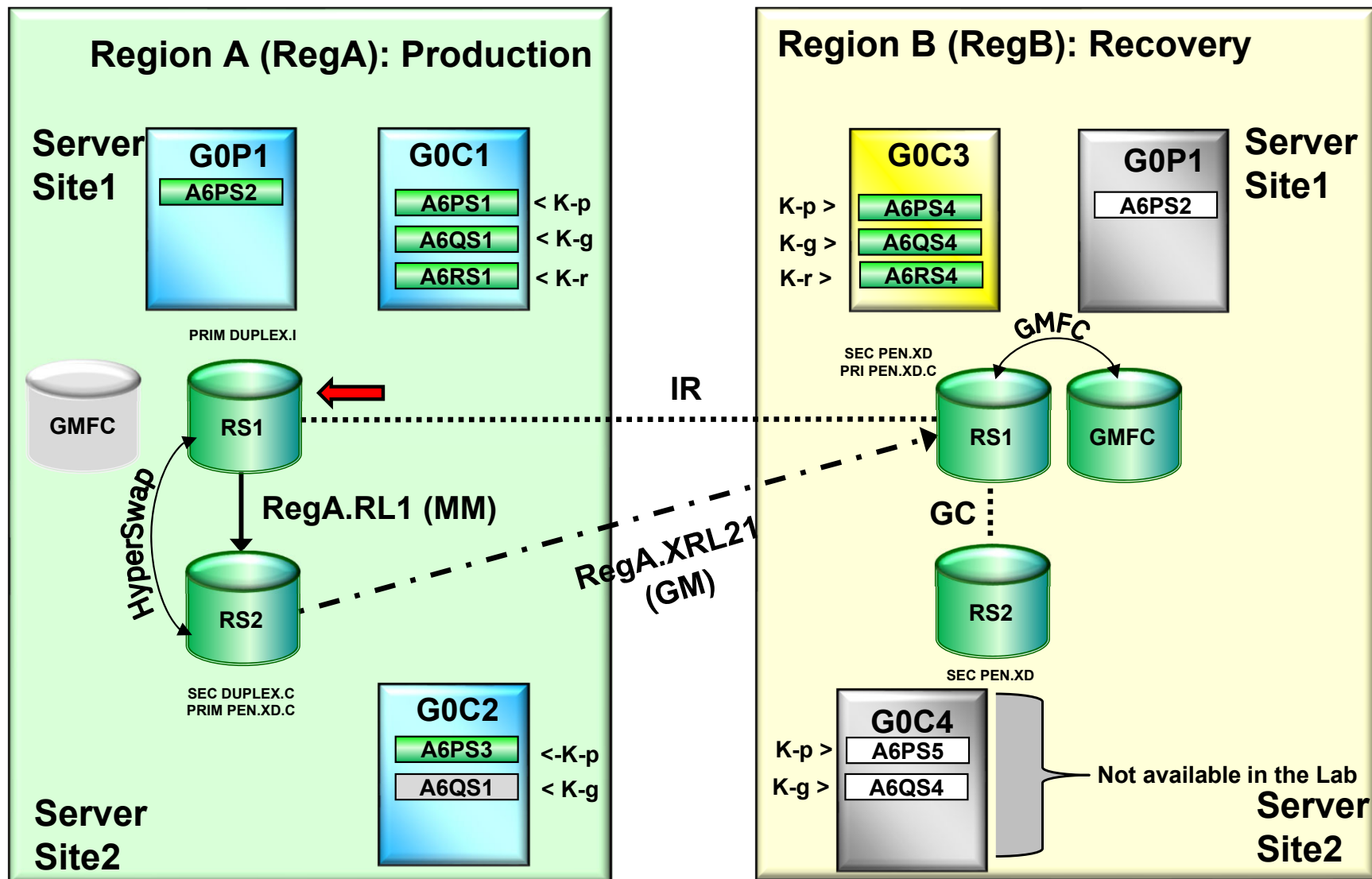
Un Planned Region Switch

- with initial configuration MGM cascaded
- Intermediate configuration



Planned Region Switch

- to initial configuration MGM cascaded
- Final configuration



Unplanned region switch – return to normal

- to initial configuration MGM cascaded

■ Used scripts:

- Control script with STOP SECONDARY
- Control script with START SECONDARY XD
- Control script with procedure MGM4SITE_PREPARE_MIRROR
- Optional Control script with procedure MGM4SITE_PREPARE_MIRROR_SWAP
- Control script with GDASD INITIALIZE
- Control script with procedure MGM4SITE_RESTORE
- Control script with procedure MGM4SITE_SWITCH

END OF LECTURE